

The total possible points are 85:

P1(20): calculation for acetylene has 10 points; using the same fashion to calculate methane worth 6 points; explaining why acetylene is chosen has 4 points.

P2(20): being able to get all the moles of the compounds worth 8 points; listing all the necessary equations with right values plugged in worth 8 points; right solution worth 4 points.

P3(25). (a) 10 points. (b) 10 points. (c) 5 points. The difference between (a) and (b) is one is a constant-pressure process and the other one is a constant volume process.

P4(20). Getting the flow specifics right worth 6 points; having the right equations and energy balance worth 10 points; reaching the right solution has 4 points.